

CS 228 : Logic in Computer Science

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So Far

- ▶ Model checking LTL
- ▶ Product of NBA and transition systems

Length of a LTL Formula $|\varphi|$

- ▶ Number of operators
- ▶ $|a| = 0$
- ▶ $|\neg a \wedge b| = |\bigcirc a \wedge b| = 2$
- ▶ $|(\bigcirc a) \cup (a \wedge \neg \neg b)| = 5$

GNBA

- ▶ Generalized NBA, a variant of NBA
- ▶ Only difference is in acceptance condition
- ▶ Acceptance condition in GNBA is a set $\mathcal{F} = \{F_1, \dots, F_k\}$, each $F_i \subseteq Q$
- ▶ An infinite run ρ is accepting in a GNBA iff

$$\forall F_i \in \mathcal{F}, \text{Inf}(\rho) \cap F_i \neq \emptyset$$

- ▶ Note that when $\mathcal{F} = \emptyset$, all infinite runs are accepting
- ▶ GNBA and NBA are equivalent in expressive power. (why?)

Word View (On the board)

- ▶ $\varphi = a \mathbf{U}(\neg a \wedge b)$
- ▶ Subformulae of $\varphi = \{a, \neg a, b, \neg a \wedge b, \varphi\}$
- ▶ Parse trees to compute all subformulae
- ▶ $w = \{a\}\{a, b\}\{\} \dots,$

Closure of φ : $cl(\varphi)$

- ▶ $cl(\varphi)$ =all subformulae of φ and their negations, identifying $\neg\neg\psi$ to be ψ .
- ▶ Example for $\varphi = a \vee (\neg a \wedge b)$
- ▶ $cl(\varphi) = \{a, \neg a, b, \neg b, \neg a \wedge b, \neg(\neg a \wedge b), \varphi, \neg\varphi\}$
- ▶ $|cl(\varphi)| \in \mathcal{O}(|\varphi|)$

Elementary Sets

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B is Propositionally Consistent

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- ▶ $\varphi_1 \wedge \varphi_2 \in B \Leftrightarrow \varphi_1 \in B \wedge \varphi_2 \in B$
- ▶ $\psi \in B \Rightarrow \neg\psi \notin B$
- ▶ $true \in cl(\varphi) \Rightarrow true \in B$

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B is Locally Consistent wrt $\cup$

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- ▶ $\varphi_2 \in B \Rightarrow \varphi_1 \cup \varphi_2 \in B$
- ▶ $\varphi_1 \cup \varphi_2 \in B, \varphi_2 \notin B \Rightarrow \varphi_1 \in B$

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B is maximal

For all $\psi \in cl(\varphi)$, $\psi \notin B \Leftrightarrow \neg\psi \in B$

Elementary Sets

Let φ be an LTL formula, and let $B \subseteq cl(\varphi)$.

B is elementary if it is propositionally, locally consistent and maximal.

- ▶ A set $B \subseteq cl(\varphi)$ is elementary if B is the set of all formulae in $\kappa \in cl(\varphi)$ with $\pi \models \kappa$ for any infinite word $\pi = A_0 A_1 \dots$.
- ▶ For this, B cannot contain contradictions : $\forall i$, either we have $a \in AP$ inside A_i or we do not have $a \in AP$ inside A_i .
- ▶ Likewise, B must respect the until semantics
- ▶ Since for any word π and any formula κ , $\pi \models \kappa$ or $\pi \models \neg \kappa$, B is maximal

Check Elementary

Let $\varphi = a \cup (\neg a \wedge b)$, $cl(\varphi) = \{a, \neg a, b, \neg b, \neg a \wedge b, \neg(\neg a \wedge b), \varphi, \neg\varphi\}$

► $B_1 = \{a, b, \neg a \wedge b, \varphi\}$

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- ▶ $B_2 = \{\neg a, \neg b, \neg a \wedge \neg b, \varphi\}$ B_2 is not a subset of $cl(\varphi)$
- ▶ $B_3 = \{\neg a, b, \neg a \wedge b, \neg\varphi\}$

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- ▶ $B_4 = \{\neg a, \neg b, \neg(\neg a \wedge b), \neg\varphi\}$ Yes, elementary

LTL φ to GNBA G_φ

- ▶ States of G_φ are elementary sets B_i
- ▶ For a word $w = A_0A_1A_2\dots$ the sequence of states $\sigma = B_0B_1B_2\dots$ will be a run for w
- ▶ σ will be accepting iff $w \models \varphi$ iff $\varphi \in B_0$
- ▶ In general, a run $B_iB_{i+1}\dots$ for $A_iA_{i+1}\dots$ is accepting iff $A_iA_{i+1}\dots \models \psi$ for all $\psi \in B_i$.

LTL to GNBA

- ▶ Let $\varphi = \bigcirc a$.
- ▶ Subformulae of φ : $\{a, \bigcirc a\}$. Let $A = \{a, \bigcirc a, \neg a, \neg \bigcirc a\}$.
- ▶ Possibilities at each state
 - ▶ $\{a, \bigcirc a\}$
 - ▶ $\{\neg a, \bigcirc a\}$
 - ▶ $\{a, \neg \bigcirc a\}$
 - ▶ $\{\neg a, \neg \bigcirc a\}$
- ▶ Our initial state(s) must guarantee truth of $\bigcirc a$. Thus, initial states: $\{a, \bigcirc a\}$ and $\{\neg a, \bigcirc a\}$

LTL to GNBA

$\{a, \bigcirc a\}$

$\{a, \neg \bigcirc a\}$

$\{\neg a, \bigcirc a\}$

$\{\neg a, \neg \bigcirc a\}$

LTL to GNBA

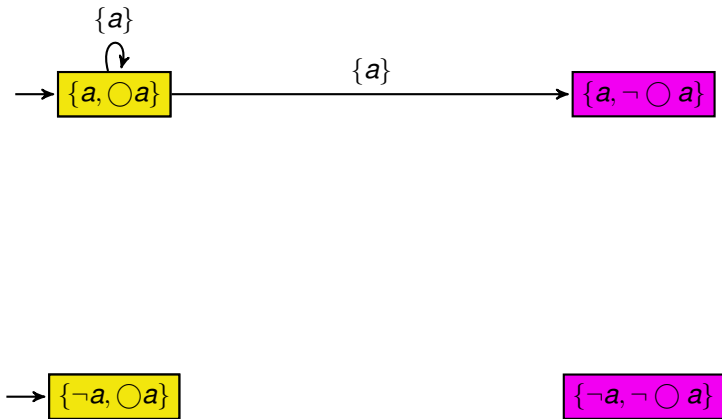
$$\rightarrow \boxed{\{a, \bigcirc a\}}$$

$$\boxed{\{a, \neg \bigcirc a\}}$$

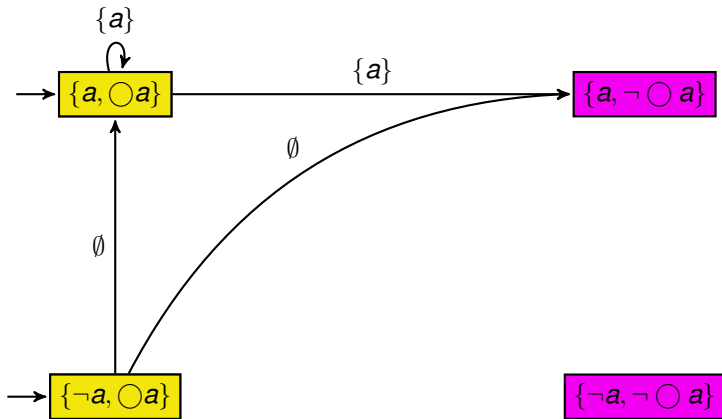
$$\rightarrow \boxed{\{\neg a, \bigcirc a\}}$$

$$\boxed{\{\neg a, \neg \bigcirc a\}}$$

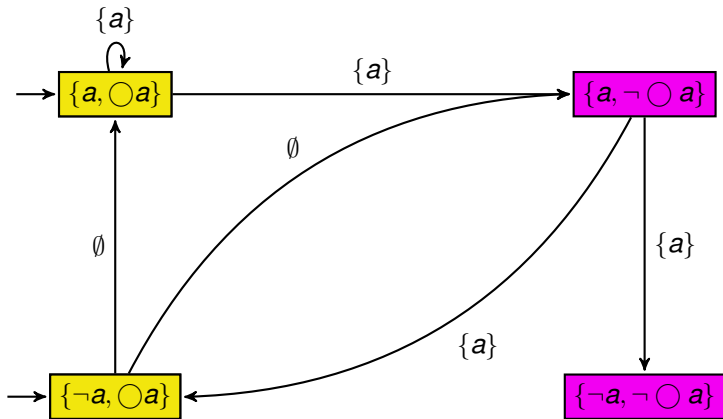
LTL to GNBA



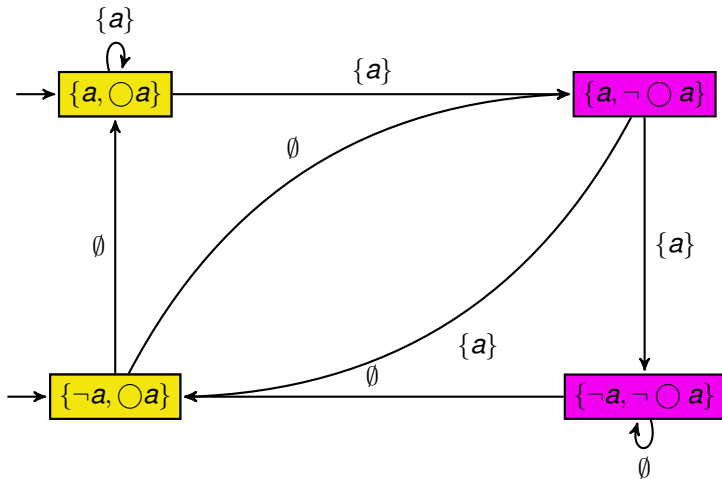
LTL to GNBA



LTL to GNBA



LTL to GNBA



LTL to GNBA

- ▶ Claim : Runs from a state labelled set B indeed satisfy B
- ▶ No good states. All words having a run from a start state are accepted.
- ▶ Correct by construction!
- ▶ Automaton for $\neg \bigcirc a$ same, except for the start states.